Structure of Computer Systems

Anca Hangan Anca.Hangan@cs.utcluj.ro

Series A (RO and EN)

Topics covered in lectures and practical applications

- Design of computer components and systems
- Study of methods used to increase the speed and efficiency of computer systems
- Study of advanced computer architectures

Passing requirements

- Laboratory and project attendance: 100% + grade ≥5
- Minimum grade of 5 in the exam to pass
- Total:
  - 60% lecture
  - 20% laboratory
  - 20% project
  - **No default points are awarded in the exam!**
  - Bonus: attendance, lecture activity
- Teams: eljeqhj

Practical activities

- Moodle
  - Attendance
  - Grades
  - Teaching materials
- Assessment
  - Laboratory: two intermediate tests (3p+3p), final lab exam (4p)
  - Project: intermediate grades (2p), documentation grade (2p), final presentation grade (6p)
- Make-up sessions:
  - 2 labs, 1 project session during the semester
  - During the resit session

AI Policy

- **Tests, exam**: **no AI**
- **Laboratory**: **AI use for assistance and as a copilot**
  - Some parts of the code may be written with AI and will be marked as such
  - The student is responsible for the architecture, for writing part of the code, and for fully understanding the parts written by AI
  - Parts written by AI are marked as such
- **Project**: **AI use for assistance and as a copilot**
  - Some parts of the code and documentation may be written with AI and will be marked as such
  - The student is responsible for the architecture, for writing part of the code, and for fully understanding the parts written by AI
  - Figures in the documentation are made by the student
  - The documentation will contain a chapter explaining where and how AI was used in developing the project and the documentation

Bibliography

- J. Hennessy , D. Patterson, Computer Architecture: A Quantitative Approach, 6th edition, 2019
- W. Stallings, Computer Organization and Architecture, 8th edition, Prentice Hall, 2010
- Zs. István, K. Kara, D. Sidler. FPGA-Accelerated Analytics: From Single Nodes to Clusters. Now Publishers Foundations and Trends in Databases, 2020
- Any book on computer architecture, microprocessors, microcontrollers or digital signal processors

Course content

- Performance of computer systems
- Design of the Arithmetic Logic Unit (ALU)
- Design strategies for the central processing unit (CPU)
- Data processing acceleration
- Interconnection systems
- Input/output interfaces
- Memory design

Computer architecture

- A trade-off between performance parameters
- Depends on the purpose and type of the computer
- Types of computers (by purpose):
  - General-purpose computers
    - clusters
    - personal computers
    - mobile computers
  - Dedicated-purpose computers
    - for scientific computing
    - military computers (safety-critical and highly reliable)
    - industrial control and automation (embedded systems)
    - measurement and analysis (e.g., medical devices, smart sensors)
- Old classification:
  - mainframes - e.g. IBM 360/370, Felix 256
  - minicomputers - PDP11, SUN workstation, Independent, Cora
  - microcomputers - microprocessor-based computers (e.g. PC, home computers)

Classification based on the number of CPUs

- Single-processor computer
- Multiprocessor computers:
  - parallel systems
    - multi-core
    - symmetric and asymmetric processors
  - distributed systems
    - personal computers and network communication for a specific (common) purpose
    - GRIDs: batch programs executed on a distributed platform; several institutions share resources
    - Cloud:
      - computer as a service
      - storage as a service
      - platform as a service
      - software as a service
      - infrastructure as a service

Classes of computers [Hennessy, Patterson]

- **IoT/Embedded computers**
  - Price: 10-100 000 EUR
  - Microprocessor price: 1-100 EUR
  - Design issues: price, energy, size, application-specific performance
- **Personal mobile devices**
  - Price: 100-1000 EUR
  - Microprocessor price: 10-100 EUR
  - Design issues: price, energy, media performance, response time
- **Desktop computers**
  - Price: 300-2500 EUR
  - Microprocessor price: 50-500 EUR
  - Design issues: price-performance, energy, graphics performance
- **Servers**
  - Price: 5000-10 M EUR
  - Microprocessor price: 200-2000 EUR
  - Design issues: throughput, availability, scalability, energy
- **Clusters/Supercomputers**
  - Price: 100 000-200 M EUR
  - Microprocessor price: > 50 EUR
  - Design issues: price-performance, throughput, energy

HPC – high performance computers

- “Highly” parallel computers – 1,024 – 1,500,000 cores or processors
- Use: scientific computing (physics, astronomy, bioinformatics, chemistry), simulation (fluid flow, weather), cryptography
- Performance: 1-20,000 Tflops
- Memory capacity: 1-700 TBytes
- Communication: InfiniBand (2-300 Gbs), Cray Gemini
- Power consumption: 10KW- 10MW (Mariselu power station ~200MW)
- Price: hard to say
- See the top 500 supercomputers (https://top500.org/lists/hpcg/2024/06/)
  - 1: Fugaku/Japan, 7,630,848 cores, 30 MW
  - 2: Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE DOE/SC/Oak Ridge National Laboratory US, 8,699,904 cores
  - 3: Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel, DOE/SC/Argonne National Laboratory US, 9,264,128 cores

HPC – high performance computers



- HPC at CERN
  - architecture: GRID
  - organization: 3 tiers
  - At least 100,000 processors in 32 countries
  - Serving over 5000 researchers
  - at UTCN: 128 quad-core processors, 512 cores





- IBM Blue Gene
  - 65,536 dual-core processors
  - Peak speed: 360 teraflops

CG-UTCN – the UTCN GRID Center

- 64 processor boards
- 128 quad-core processors,
- 512 cores
- 1024 virtual processors (hyper-threading)
- storage: 12 Tbytes
- price: 2,000,000 RON



CloudUT

- 10 compute nodes (320 cores), 5120 GB RAM, 167 TB storage capacity, RAID 5;
- 3 GPU nodes with graphics accelerators: (node1) 2xGPU Tesla V100, 32 GB dedicated memory; (node2) 2xGPU Tesla V100, 32 GB + 1xGPU NVIDIA A100, 40 GB; (node3) 4xGPU NVIDIA A40, 48GB, support for Q- and C-type vGPU, cards interconnected via NVIDIA NVLink 112.5 GB/s;
- 25Gbps internal connectivity and 10Gbps external connectivity;
- VMware with support for management, virtualization, orchestration and automation, with vCloud capabilities.



Classes of parallelism

- **Application-level parallelism:**
  - Data-level parallelism
  - Task-level parallelism
- **Hardware-level parallelism:**
  - Instruction level: pipelining, speculative execution
  - Vector architectures, GPUs, multimedia instruction sets: a single instruction applied to a collection of data
  - Thread level: a lot of interaction between threads
  - Request level: very little interaction between threads

Parallel architectures (Flynn's taxonomy)

- SISD – uniprocessor
- SIMD – data-level parallelism (vector architectures, GPUs, etc.)
- MISD – pipeline (?)
- MIMD – multiprocessor, task-level parallelism

Functional requirements for hardware

- **Application domain**:
  - **IoT**: real-time constraints, power limitations
  - **Mobile**: soft real-time performance, interactive applications, energy efficiency
  - **Desktop**: balanced performance for diverse tasks, interactive applications
  - **Servers**: database support, transaction processing, reliability, availability, scalability
  - **Clusters**: throughput performance for independent tasks, memory error tolerance, energy consumption proportional to the amount of useful work done
- **Software compatibility**: compilers, HW optimizations
- **OS requirements**: address space size, memory management and protection
- **Standards**: FP format, I/O interfaces, OS, networks, programming languages (affect the instruction set)

Functional requirements for hardware

- Execution time
- Response time
- Bandwidth
- Memory capacity
- Input/output interfaces
- Development (e.g. OS, programming environments)
- Size and form factor
- Predictability, safety and fault/error tolerance
- Scalability
- Costs: absolute and relative

Performance metrics

- Bandwidth or throughput:
  - The total amount of work done in a given time interval
  - Mbytes/s for a disk transfer
- Latency or response time
  - The time between the start and the completion of an event
  - Milliseconds for a disk transfer
- Capacity – more important for memories and storage units
- Energy efficiency
  - Performance per watt
  - HPC FLOPS/W, microcontrollers MIPS/W, WSNs J/bit

Execution time

- Arithmetic operation:
  - e.g., multiplication is 30-40 times slower than addition
  - single or multiple clock periods

- Instruction:
  - simple and complex instructions have different execution times
  - **average execution time = $\Sigma\, t_{instruction}(i) * p_{instruction}(i)$**, where $p_{instruction}(i)$ – the probability of instruction "i"
  - Predictable/non-predictable systems – with fixed instruction execution time/with variable execution time

Execution time

- Procedure, task: the time needed to perform a given function (e.g., sorting, printing, selection, I/O operations, context switching)
- Transaction: execution of a sequence of operations for accessing and modifying a database
- Application: 3D rendering, fluid flow simulation, statistical data computation

Response time

- Solutions:
  - Best effort – batch programming – reaction time does not matter
  - Interactive systems – event-driven systems – reaction time matters but is not guaranteed
  - Real-time systems – the worst-case execution time (WCET) is guaranteed
- Scheduling strategies for single-processor or multiprocessor systems
- Influencing factors:
  - execution time of interrupt routines or procedures
  - time needed for context switching
  - background execution of operating system threads

Memory capacity and latency

- Cache memory: SRAM, very low latency (0.5-10ns), small capacity (MB)
- Internal memory: SRAM or DRAM, medium latency (100ns), medium capacity (GB)
- External memory (storage): SSD (16us), HDD (2ms), Flash - very large capacity (TB)

Input/output interfaces

- Very diverse or dedicated to a specific purpose
- Input devices: keyboard, mouse, video camera, microphone or sensors/transducers
- Output devices: printer, video, sound, or actuators
- Input/output: storage devices

Development requirements

- Services provided by the operating system (e.g.: application execution, display, communication, file system, etc.)
- Programming and debugging environments
- Development kits (minimal hardware and software for building dedicated systems)

Size and form factor

- Supercomputers – minimal size restrictions
- Personal computers – desktop, laptop, tablet PC – some limitations
- Mobile devices – phones, "wearable devices", medical devices – significant size and form restrictions
- Dedicated systems – significant size and form restrictions

Predictability, safety and fault tolerance

- Predictable execution time
- Controllable, guaranteed quality and safety
- Safety-critical systems, industrial computers, medical devices

Cost

- Absolute or relative (cost/performance, cost/bit)
- Cost restrictions for embedded systems